

Variational Neural Cellular Automate

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VNCA Model Structure

Variational autoencoder with classical encoder and neural cellular automata decoder

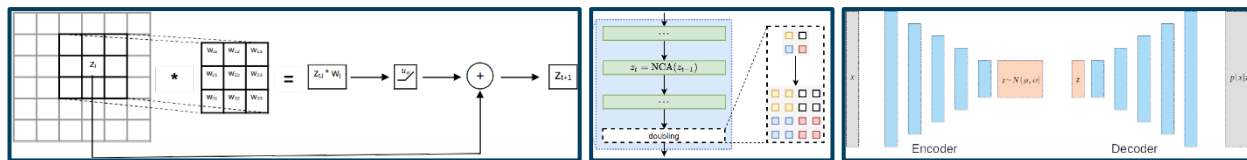


Fig 1. NCA Single Layer

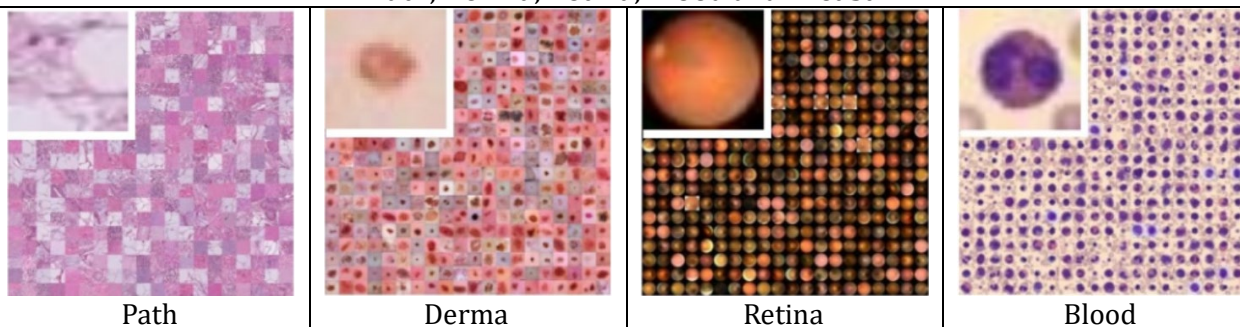
Fig 2. Doubling NCA

Fig 3. Variational NCA

- VNCA combines NCA and (Variational Autoencoders)VAE. It processes states layer by layer using local update rules, evolving the initial state to a high-resolution state. (Fig 2)
- The model updates each cell's state through conv operations and a learnable non-linear function. (Fig 1)
- It encodes the input into latent variable via variational inference and reconstructs through a decoder. (Fig 3)
- VNCA maximizes the Evidence Lower Bound (ELBO), optimizing both the KL divergence and reconstruction error.

MedMNIST - Dataset

We use colored (RGB) 2D dataset from MedMNIST for our training of models. Used datasets are: **Path, Derma, Retina, Blood and Breast.**



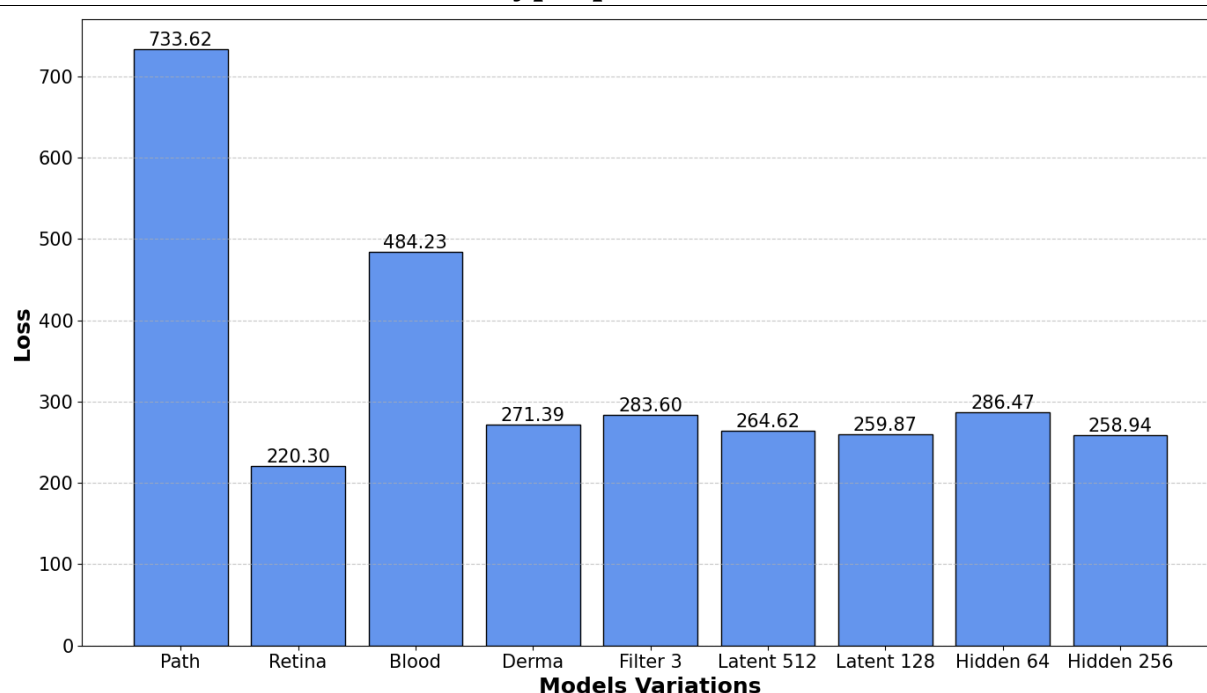
Path

Derma

Retina

Blood

Model & Hyperparameter Losses



4 models trained with defaults setting. 5 models trained for Hyperparameter tune. The default parameters are Filter Size = 5, Latent Space = 256, Hidden State = 128, Epochs = 50K

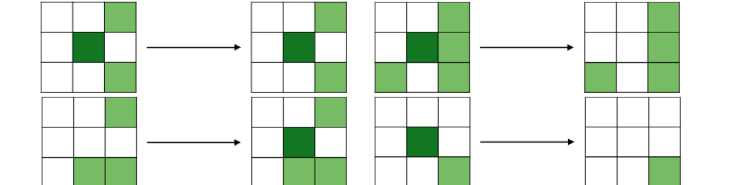
Loss Type: Evidence Lower Bound, consisting of a reconstruction error and a KL divergence term.

Losses Explanation: The extrem high Path and Blood losses may due to instability during training caused by gradient vanish. They ended unexpected during training and then resumed from last saved epoch.

Zhiqian Yu - Coding & Training, Dieter Meier Helge - Coding & Training,
Xiaoyan Xue - Coding & Inference & Report, Yubao Ma - Report, All Member - Poster

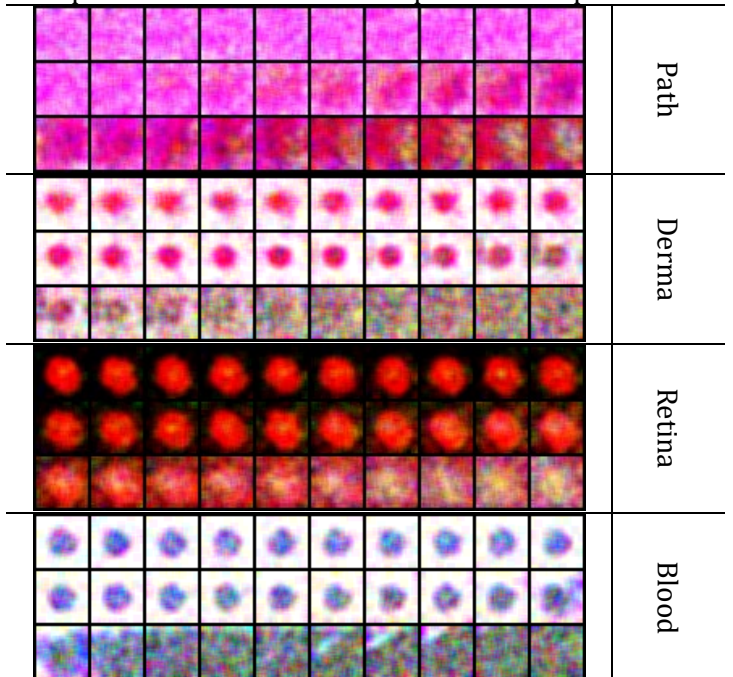
Conways Game of Life

Survival, death and birth of cell depending on neighbours



MedMNIST Training Results

With all the datasets we use, we train them with 50K epoch. Results are the 30 Steps evolution process.



Path

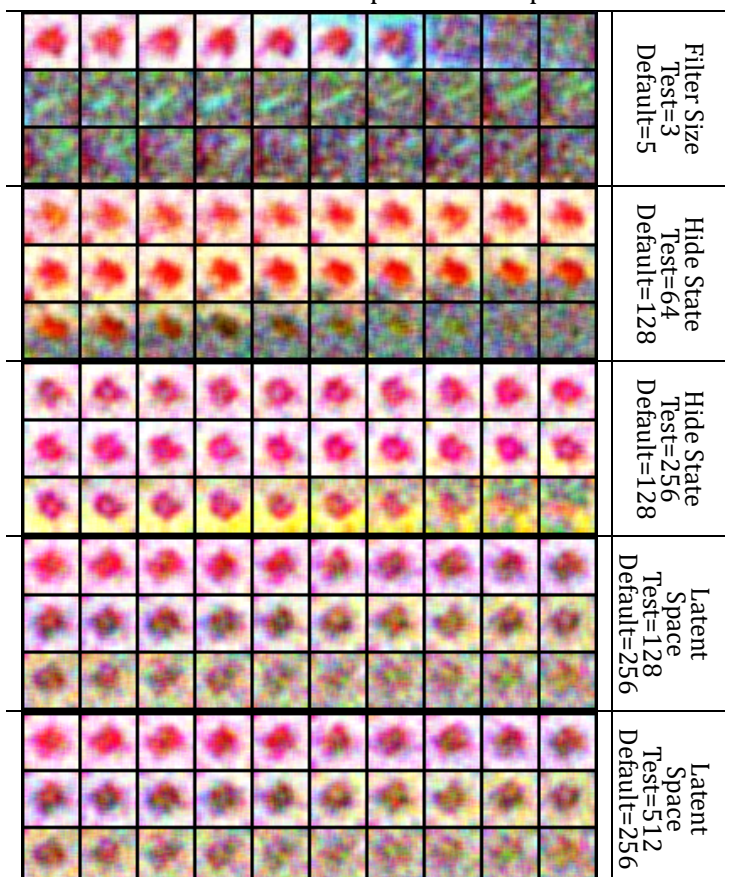
Derma

Retina

Blood

Hyperparameter Tune

We choose Derma dataset for Hyperparameter tests, with 50K epochs and 1 variable changed each time. Results are the 30 Steps evolution process.



Filter Size
Test=3
Default=5

Hide State
Test=64
Default=128

Hide State
Test=256
Default=128

Latent
Space
Test=128
Default=256

Latent
Space
Test=512
Default=256